

Commerce XI Mathematics & Statistics · Part 2

Unit 4: Bivariate Frequency Distribution and Chi Square Statistic

A quick reference based on Unit 4, textbook pages 45-55.

How to use this guide

Use	What to check
Before solving	Identify the two variables, their classes or categories, the observed frequencies, and the required table or statistic.
During solving	Tally each pair once, check row and column totals, and calculate expected frequencies before the chi-square contributions.
Before answering	Verify that both marginal totals equal N, expected totals agree with observed totals, and chi-square is non-negative.

Scope

Bivariate frequency distribution · marginal and conditional distributions · categorical variables · contingency tables · expected frequencies · chi-square statistic

Textbook Answer Notes

Section 4.2, pages 45-47: the sentence stating that the first observation is (16, 150) is incorrect. The original data show the first observation is (33, 250); (16, 150) is the seventh observation. The tables are consistent with the original 15 data pairs.
Exercise 4.1, Q4: the question lists 16 pairs, but the printed answer table totals 15 and omits the final pair (26, 21). The App uses all 16 printed pairs.

1. Bivariate Frequency Distribution

A bivariate frequency distribution records two variables observed on the same population. Each observation is an ordered pair (x, y). Place X classes across columns, Y classes across rows, and tally every pair in its corresponding cell.

Essential notation

Symbol	Meaning
x_i	ith value or class of X
y_j	jth value or class of Y
f_{ij}	frequency of the pair (x_i, y_j)
N	grand total of all frequencies

Marginal distributions

Column totals with X give the marginal frequency distribution of X. Row totals with Y give the marginal frequency distribution of Y. Both sets of totals must sum to N.

Conditional distributions

A conditional distribution shows one variable after fixing a value or class of the other. Read one selected row for X given Y, or one selected column for Y given X.

2. Categorical Variables and Contingency Tables

Categorical variables take non-numerical values, such as gender, blood group, mother tongue, preference or outcome. A two-way frequency table for two categorical variables is called a contingency table.

3. Chi-square Statistic

The chi-square statistic measures the difference between observed and expected cell frequencies in a contingency table.

$$\chi^2 = \sum [(O_{ij} - E_{ij})^2 / E_{ij}]$$

$$E_{ij} = (R_i \times C_j) / N$$

Symbol	Meaning
O_{ij}	observed frequency

E_{ij}	expected frequency
R_i	row total
C_j	column total
N	grand total

Calculation method

1. Complete row totals, column totals and N .
2. Calculate each expected frequency using $R_i C_j / N$.
3. For every cell calculate $(O - E)^2 / E$.
4. Add all contributions.
5. Check that χ^2 is non-negative.

Worked check: handedness example

For men/women and right/left handedness, the observed table has row totals 80 and 120, column totals 75 and 125, and $N = 200$. Expected frequencies are 30, 50, 45 and 75. The four contributions are approximately 3.33, 2.00, 2.22 and 1.33, giving $\chi^2 = 8.88$.

Worked check: colour-preference example

For boys/girls choosing pink, blue or orange, both row totals are 100 and column totals are 68, 108 and 24. Expected frequencies in each row are 34, 54 and 12. The resulting χ^2 is 6.55.

4. Common Checks

- Every ordered pair must be tallied exactly once.
- Row totals and column totals must both add to N .
- Expected frequencies in a row and column preserve the corresponding marginal totals.
- Use frequencies, not percentages, in the chi-square calculation unless frequencies are first recovered.
- Do not round intermediate expected frequencies too early.

5. Textbook Answer Notes

Section 4.2, pages 45–47: The explanatory sentence on page 45 incorrectly states that the first observation is (16, 150). The original 15-day data show the first observation is (33, 250), and the following sentence correctly places the tally at (33, 250). The pair (16, 150) is the seventh observation. Tables 4.2–4.7 are consistent with the original 15 data pairs.

Exercise 4.1, Q4: The question lists 16 paired observations. The printed answer table totals only 15 and omits the final observation, $(X, Y) = (26, 21)$. Classifying all 16 printed pairs gives row totals 5, 7, 4 and column totals 7, 5, 4. The App uses the complete 16-pair result.

Exercise 4.2 printed answers: $Q_1 = 0$, $Q_2 = 3.37$, $Q_3 = 1.07$, $Q_4 = 2.66$ and $Q_5 = 30.625$. Miscellaneous Exercise 4 printed numerical answers: $Q_6 = 2.38$, $Q_7 = 0.67$, $Q_8 = 10.614$, $Q_9 = 4.31$ and $Q_{10} = 0.4076$.